

Marta Marchiori Manerba *(she/her)*

2° Year PhD Student in AI for Society

About

Marta Marchiori Manerba (she/her) is a Graduate Student in Digital Humanities at the University of Pisa and currently a Ph.D. student in AI. During her studies, she explored the relationship between technology and human rights. She works on Fairness and Explainability in Natural Language Processing, focusing on digital discrimination and algorithmic biases. She holds a Bachelor's Degree in Digital Humanities from the University of Pisa, during which she developed a strong interest in hate speech detection towards minorities in online discourse.

Contacts

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Research Interests

NLP; Responsible Language Technologies;
Human-Centered AI; Explainability; Transparency;
Fairness; Algorithmic Auditing; ML Evaluation;
Data Awareness; Intersectionality; Digital
Discrimination; Abusive Language Detection

Education

NOVEMBER 2021 - ONGOING

National PhD in AI for Society - [Computer Science Department](#), University of Pisa

Supervisors: Dr. Riccardo Guidotti; Prof. Salvatore Ruggieri

Doctoral fellowship on "*Science and technology of interpretable machine learning and explanation of AI-assisted decision making*"

FEBRUARY 2019 - JULY 2021

Master's Degree - [Digital Humanities: Language Technologies](#), University of Pisa

Supervisor: Dr. Riccardo Guidotti. Mark: 110/110 with honors

Thesis project on Fairness Auditing in Abusive Language Detection Systems carried out in the field of *Explainability and Fairness in ML*

OCTOBER 2015 - FEBRUARY 2019

Bachelor's Degree - [Digital Humanities](#), University of Pisa

Supervisor: Dr. Maria Claudia Buzzi. Mark: 109/110

Thesis project carried out in the field of *Assistive Technology for Education*: development of a website for the research of tools and ICT resources to support cognitive training

Research and Academic Experience

JANUARY-JULY 2022

Supervisor of a Master Thesis - [Digital Humanities](#), University of Pisa

The contribution, titled "Genetic Fairness-Enhancing Data Generation Framework", was supervised alongside Riccardo Guidotti and Martina Cinquini

2022

Reviewer - LREC Workshop on Perspectivist Approaches to NLP ([NLPerspectives](#))

20-22 JANUARY 2021

Assistant moderator - Association for Digital Humanities and Digital Culture Conference ([AIUCD2021](#))

OCTOBER - DECEMBER 2020

Traineeship - Digital Humanities group, Fondazione Bruno Kessler

Fairness analysis of abusive language detection systems detecting unintended models biases with CheckList: creation of synthetic linguistic data to test a range of social prejudices (e.g. sexism, racism and ableism)

OCTOBER - DECEMBER 2020

Academic Tutor of *Programming and Theoretical Fundamentals* - Digital Humanities, University of Pisa
Assistance during the laboratory: preparing exercises and assignments, helping students with coding during classes, clarifying basic constructs and concept in Javascript

MAY - SEPTEMBER 2018

Traineeship in *Assistive Technology for Education* - Institute of Computer Science and Telematics, CNR
Mapping and categorization of ICT resources to support cognitive training through the construction of an inventory of tools available on the internet; Implementation of a digital catalogue to make the collection usable

Publications

Conference Publications

In-person presentation

Marta Marchiori Manerba and Riccardo Guidotti. "Investigating Debiasing Effects on Classification and Explainability". AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES), 2022

Online presentation, Best Paper Award

Marta Marchiori Manerba and Riccardo Guidotti. "FairShades: Fairness Auditing via Explainability in Abusive Language Detection Systems". IEEE Conference on Cognitive Machine Intelligence (CogMI), 2021

Workshop Publications

In-person presentation

Marta Marchiori Manerba and Virginia Morini. "Exposing Racial Dialect Bias in Abusive Language Detection: Can Explainability Play a Role?". ECML PKDD International Workshop on eXplainable Knowledge Discovery in Data Mining (XKDD), 2022

In-person presentation

Marta Marchiori Manerba, Riccardo Guidotti, Lucia Passaro, Salvatore Ruggieri. "Bias Discovery Within Human Raters: A Case Study of the Jigsaw Dataset". LREC Workshop on Perspectivist Approaches to NLP (NLPerspectives), 2022

Arianna Muti, Marta Marchiori Manerba, Katerina Korre, and Alberto Barrón-Cedeño. "LeaningTower@LT-EDI-ACL2022: When Hope and Hate Collide". ACL Workshop on Language Technology for Equality, Diversity and Inclusion (LT-EDI), 2022

Online presentation

Marta Marchiori Manerba and Sara Tonelli. "Fine-Grained Fairness Analysis of Abusive Language Detection Systems with CheckList". Workshop on Online Abuse and Harms (WOAH!), 2021

Currently under review

Federico Mazzoni, Marta Marchiori Manerba, Martina Cinquini, Riccardo Guidotti, and Salvatore Ruggieri. "FairGen: A Genetic Fairness-Enhancing Data Generation Framework". Intelligent Data Analysis (IDA), 2023

Research Dissemination: Engagements & Talks

2022-2023

Organisation of a panel discussion titled "Picture a Scientist" - Computer Science Department, University of Pisa

Won funds for PhD students' scientific initiatives. The event aims to analyse the reasons and factors behind the lower percentage of women choosing to pursue education and careers in STEM fields

MAY 2022

High school civic education lesson titled "Ethics in AI Systems: Biases in ML Models" - IIS Cobiانchi in Verbania, Italy

MARCH 2022

Video for university choice orientation on Digital Humanities and the impacts of AI - "A Choice for Life"

MARCH 2022

University orientation event for high schools: talk titled "Ethics in AI Systems and the Timnit Gebru case" - Computer Science Department, University of Pisa

Digital Advocacy

NOVEMBER 2022 - IN PROGRESS

Team member - Privacy Network

Non-profit association that promotes a culture of privacy and responsible use of technology and that advocates for digital rights

NOVEMBER 2019 - IN PROGRESS

Vice president and board member (2022) - KRINO

Cultural association that organizes panels, workshops and events within the field of Digital Humanities, with the aim of integrating and contaminating sciences and arts

- Active participation in projects of critical divulgation of topics such as: computational thinking, datafication, digital culture, demystification of AI systems, digital art, live coding
 - Collaborative and collective writing and editing of posts, articles, newsletters
 - Internal training sessions
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Memberships

2020 - IN PROGRESS

Algorithmic Justice League

FEBRUARY 2021 - DECEMBER 2021

Associazione Italiana di Linguistica Computazionale (Italian Association of Computational Linguistics)

JANUARY 2021 - DECEMBER 2021

Associazione per l'Informatica Umanistica e la Cultura Digitale (Association for Digital Humanities and Digital Culture)

Volunteering

SEPTEMBER 2017 - OCTOBER 2022

Scout educator - Association of Italian Guides and Scouts (AGESCI)

Association contributing to the education and personal development of youth according to scouting principles and methods

- Organisational and coordination skills; ability to work in teams and on projects
- Communication and relational skills developed as educator; aptitude for listening and confrontation